

to adopt automation. Besides, the market (for manufacturers) is not just India. You can't compete with other manufacturing destinations with legacy machines and tools."

Also, there's not much value in making basic components like nuts and bolts. Gokul N.A., founder of CynLr, a Bengaluru-based robotics company, says, "Everything we ship globally should have a piece of software or intellectual property (IP) inside it."

That won't come in legacy units. For instance, India makes smart LED bulbs, but what makes them smart is the chip, which is imported. This will change in the coming years as the semiconductor units being built in India start churning out chips by 2027-28.

TRANSFORMING INDUSTRY

Indeed, inside many of the new factories coming up across India's electronics clusters—from Noida and Chennai to Bengaluru—the shop floor is quietly changing. For instance, Bengaluru-based space tech company GalaxEye uses Dassault Systèmes' engineering software to create digital twins of satellites before building them. This enables extensive virtual testing and simulation of complex space hardware before production starts. JSW Motors also uses digital twins to create simulations of electric and plug-in hybrid vehicles. Gokul says, "In a digital twin environment,



"Penetration gap (of robots) compared to global levels is still very wide. India's robot density is sixth, behind Germany, South Korea, US, Japan and China"

GOKUL NA
Co-founder, CynLr



"When companies build new plants today, they often want to build every component in a new plant digitally with the same physical properties as the real equipment"

DEEPAK NG
Managing Director,
Dassault Systèmes India

companies can test and fail repeatedly. Digital twins help augment and accelerate decision-making before the final product is made."

At the heart of this transformation is the growing ability of machines to communicate with each other. "AI is changing the landscape of manufacturing—with machines on assembly lines now speaking to each other," says Gujral. As data collection speeds

up across machines and production lines, AI systems can analyse that information in real time and recommend adjustments that improve efficiency.

This constant stream of data allows factories to detect problems before they occur. Predictive maintenance systems, powered by AI models, can forecast when an equipment is likely to fail and schedule repairs be-

+ Virtual twins help plan how a factory will look, even before it is built

+ Machines talk to each other, making repair, maintenance, workflow much smarter and easier

+ IoT sensors collect every data and rely to dashboards for analytics